Assignment No	**2**
Title	**Analytical Queries**
Objective	**Roll_Up, CUBE , First, Last, Lead, Lag, Row_number ,Rank and Dense Rank**
Roll No	**MCA2649**

Source Code:-

```
CREATE TABLE Employee (
   empNo    NUMBER(6),
   Name     VARCHAR2(10),
   Position VARCHAR2(6),
   manager  NUMBER(6),
   JoinDate DATE,
   Salary   NUMBER(7),
   DeptNo   NUMBER(3)
);
```

Output:

Creating a Table: Employee Table and Inserting Values in the Table

```
SQL> CREATE TABLE Employee (
  2       empNo    NUMBER(6),
  3       Name     VARCHAR2(10),
  4       Position VARCHAR2(6),
  5       manager  NUMBER(6),
  6       JoinDate DATE,
  7       Salary   NUMBER(7),
  8       DeptNo   NUMBER(3)
  9  );

Table created.
```

```
SQL> INSERT INTO Employee VALUES
  2  (1632,'Sam','Asstt',131632,TO_DATE('9-6-2026','DD-MM-YYYY'),11370,102);

1 row created.

SQL> INSERT INTO Employee VALUES(1001,'SMITH','CLERK',7839,TO_DATE('10-1-2020','DD-MM-YYYY'),1200,10);

1 row created.
```

1) ROLLUP:-

- **One Column Without ROLLUP:** Displays the total salary of employees for each department without showing subtotals or a grand total.

```
SQL> SELECT DeptNo, SUM(Salary) AS total
  2  FROM Employee
  3  GROUP BY DeptNo
  4  ORDER BY DeptNo;

    DEPTNO      TOTAL
---------- ----------
        10       6700
        20       7300
        30       7600
        40       8000
       102      11370
```

- **One Column With ROLLUP:** Displays the total salary of employees for each department along with the grand total.

```
SQL> SELECT DeptNo, SUM(Salary) AS Total FROM Employee GROUP BY ROLLUP(DeptNo) ORDER BY DeptNo;

    DEPTNO      TOTAL
---------- ----------
        10       6700
        20       7300
        30       7600
        40       8000
       102      11370
                40970

6 rows selected.
```

Difference: `GROUP BY DeptNo` shows the total salary for each department, while `GROUP BY ROLLUP(DeptNo)` shows the department-wise totals **plus the grand total**.

- **Multiple Columns Without ROLLUP:** Displays the total salary for each combination of Department and Position without showing subtotals or a grand total.

```
SQL> SELECT DeptNo, Position, SUM(Salary) AS Total FROM Employee GROUP BY DeptNo, Position ORDER BY 1,2;

    DEPTNO POSITI      TOTAL
---------- ------ ----------
        10 ANALYT       3000
        10 CLERK        1200
        10 MGR          2500
        20 CLERK        1300
        20 MGR          2800
        20 STAFF        3200
        30 CLERK        1400
        30 MGR          2700
        30 STAFF        3500
        40 CLERK        1500
        40 MGR          2900

    DEPTNO POSITI      TOTAL
---------- ------ ----------
        40 STAFF        3600
       102 Asstt       11370

13 rows selected.
```

- **Multiple Columns With ROLLUP:** Displays the total salary for each Department and Position combination, along with department-wise subtotals and a grand total.

```
SQL> SELECT DeptNo, Position, SUM(Salary) AS Total
  2  FROM Employee
  3  GROUP BY ROLLUP(DeptNo, Position)
  4  ORDER BY 1,2;

    DEPTNO POSITI      TOTAL
---------- ------ ----------
        10 ANALYT       3000
        10 CLERK        1200
        10 MGR          2500
        10              6700
        20 CLERK        1300
        20 MGR          2800
        20 STAFF        3200
        20              7300
        30 CLERK        1400
        30 MGR          2700
        30 STAFF        3500

    DEPTNO POSITI      TOTAL
---------- ------ ----------
        30              7600
        40 CLERK        1500
        40 MGR          2900
        40 STAFF        3600
        40              8000
       102 Asstt       11370
       102             11370
                       40970

19 rows selected.
```

Difference: GROUP BY DeptNo, Position gives only combination-wise totals, while ROLLUP(DeptNo, Position) gives combination-wise totals **plus department subtotals and a grand total**.

2)CUBE:-

- **One Column With CUBE:** Displays the total salary of employees for each department along with the grand total.

```
SQL> SELECT DeptNo, SUM(Salary) AS Total
  2  FROM Employee
  3  GROUP BY CUBE(DeptNo)
  4  ORDER BY DeptNo;

    DEPTNO      TOTAL
---------- ----------
        10       6700
        20       7300
        30       7600
        40       8000
       102      11370
                40970
```

- **Multiple Columns With CUBE:** Displays the total salary for each Department and Position combination, along with department-wise subtotals, position-wise subtotals, and the grand total.

```
SQL> SELECT DeptNo, Position, SUM(Salary) AS Total
  2  FROM Employee
  3  GROUP BY CUBE(DeptNo, Position)
  4  ORDER BY 1,2;

    DEPTNO POSITI      TOTAL
---------- ------ ----------
        10 ANALYT       3000
        10 CLERK        1200
        10 MGR          2500
        10              6700
        20 CLERK        1300
        20 MGR          2800
        20 STAFF        3200
        20              7300
        30 CLERK        1400
        30 MGR          2700
        30 STAFF        3500

    DEPTNO POSITI      TOTAL
---------- ------ ----------
        30              7600
        40 CLERK        1500
        40 MGR          2900
        40 STAFF        3600
        40              8000
       102 Asstt       11370
       102             11370
           ANALYT       3000
           Asstt       11370
           CLERK        5400
           MGR         10900

    DEPTNO POSITI      TOTAL
---------- ------ ----------
           STAFF       10300
                       40970

24 rows selected.
```

Difference: Multiple Columns With CUBE displays Department–Position totals, department-wise subtotals, position-wise subtotals, and the grand total, whereas **Multiple Columns Without CUBE** displays only Department–Position totals.